Deep Dynamic Adaptation Network Based on Joint Correlation Alignment for Cross-Scene Hyperspectral Image Classification

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***Abstract*— Deep learning methods face significant challenges in practical cross-scene classification tasks of hyperspectral images (HSIs), primarily due to the difficulty of acquiring labels and the issue of inconsistent distribution caused by spectral drift. To tackle the above issues, we propose a deep dynamic adaptation network based on joint correlation alignment (DDAN-JCA) for cross-scene HSI classification. First, the dual-channel residual network (DCRN) and the attention mechanism module (AMM) are employed to extract spatial–spectral joint features from both the source domain (SD) and target domain (TD). Then, the method of correlation alignment (CORAL) is employed to minimize the marginal distribution discrepancy between two domains and further reduce the conditional distribution discrepancy of each class. Finally, a dynamic distribution adaptation strategy is used to dynamically adjust the importance of marginal distribution and conditional distribution by using a balance factor. DDAN-JCA can achieve unsupervised classification without using target labels. The performance of DDAN-JCA has been validated using three hyperspectral datasets, and the experimental results demonstrate that DDAN-JCA significantly enhances classification accuracy and exhibits greater robustness compared with state-of-the-art methods.**

***Index Terms*— Cross-scene classification, deep dynamic adaptation network based on joint correlation alignment (DDAN-JCA), domain adaptation (DA), hyperspectral image (HIS).**

I. Introduction

HYPERSPECTRAL images (HSI) typically capture the reflectance signals of objects in the range of 400–2500 nm with >100 bands [1], [2]. Thanks to the fine spectral resolution, HSI is widely used for agricultural monitoring, resource exploration, environmental protection, and other fields [3], [4], [5], [6], [7]. Therefore, how to obtain effective information from HSI is an important research direction.

Classification aims to obtain distribution information on ground cover from HSI data and provide data support for environmental monitoring and resource exploration. However, the classification of HSI poses significant challenges due to factors, such as the large amount of data, high dimensionality, and limited availability of labeled data. In practical applications, the HSI used for training the classification model (called source domain, SD) and the HSI to be classified (called target domain, TD) are obtained from different scenes. The problem of inconsistent distribution arises from variations in acquisition environmental conditions or sensors, significantly complicating the HSI classification process [8], [9]. To solve the above problems, domain adaptation (DA) first aligns the distribution between SD and TD to strengthen the generalization ability of the model, and transfers the SD model to TD model for classification, which reduces the dependency of TD with labels [10], [11].

Traditional DA methods include those based on distribution alignment, feature selection, and subspace learning [12], [13], [14], [15]. Distribution alignment-based methods address the distinct probability distributions between source data and target data by minimizing the distribution discrepancy through linear or nonlinear transformations. This includes aligning the marginal distribution, conditional distribution, and joint distribution [16], [17], [18]. Feature selection-based methods assume the existence of shared features between SD and TD, extracting these shared features to align the distributions for DA classification [19], [20], [21]. Subspace learning-based methods posit the existence of a subspace with similar distributions for source and target data, enabling DA classification within this shared subspace. These methods employ statistical feature alignment and manifold learning techniques for subspace extraction [22], [23]. In contrast to traditional DA methods, deep DA methods employ deep neural networks (DNNs) to automatically learn features [24]. These methods include discrepancy based and adversarial based [25]. Discrepancy-based deep DA methods emphasize the alignment of marginal or conditional distributions between different domains, and these methods commonly employ adaptation layers that utilize metrics like the maximum mean discrepancy (MMD). Adversarial-based deep DA methods employ adversarial learning to minimize cross-scene discrepancies,

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producing transferable and domain-invariant features. Deep DA methods utilize the power of deep learning to automatically extract discriminative features, enabling effective DA. These methods offer promising solutions for addressing the challenges posed by cross-scene HSI classification tasks.

This study primarily focuses on discrepancy-based deep DA methods, which have shown significant efficacy in addressing the challenges of DA. MMD [26] is a prominent distribution alignment strategy employed in numerous deep DA methods. Long and Wang [27] proposed a method called deep adaptation network (DAN) in their study. DAN utilizes DNNs to learn transferable cross-domain features. By incorporating adaptation layers based on multikernel variants of MMD (MK-MMD), DAN effectively adjusts the marginal distribution between domains. Li et al. [28] proposed a method called two-stage deep DA (TDDA). First, TDDA utilizes MMD to learn a deep embedding space, thereby minimizing the distribution discrepancy between different domains. Second, a spatial–spectral coupled network is employed to further minimize distribution shift and enhance the discriminative power of the deep embedding space. Another notable distance measurement method is correlation alignment (CORAL) [29], which quantifies the distribution distance between domains by aligning the second-order statistics (covariance) of SD and TD. And, it accomplishes this alignment without the need for target labels. Sun and Saenko [30] proposed D-CORAL, an extension of CORAL tailored for deep architectures. D-CORAL employs a nonlinear transformation to align the marginal distribution between domains, leading to enhanced performance in DA. However, the existing deep DA methods usually focus on aligning the marginal distribution and often overlook the influence of each sample in the SD and TD on the network weights. Long et al. [31] proposed a joint adaptation network (JAN), which uses labeled SD and pseudo-labeled TD to construct MMD and conditional MMD, and achieves alignment of joint distributions by means of adversarial training. Wang et al. [32] introduced a DA method specifically designed for multitemporal HSI using a graph neural network (GNN). They employed GNN for feature extraction and integrated both domain-wise CORAL and class-wise CORAL into the GNN network, enabling comprehensive adaptation performance for joint distribution alignment. Zhu et al. [33] introduced AMRAN, an attention-based multiscale residual adaptation network, tailored specifically for cross-scene classification tasks. AMRAN includes a residual adaptation module to align marginal distribution, an attention module for robust feature extraction, and a multiscale adaptation module for extracting features at different scales and aligning conditional distribution. The integration of these components led to significant improvements in the cross-scene classification performance. In summary, the studied deep DA methods have provided compelling evidence for the advanced and effective utilization of deep learning in DA.

Traditional manual feature extraction methods often make assumptions about specific data distributions in HSI and extract features from bands tailored to particular problems, potentially resulting in inaccurate data representations and limited applicability to complex scenarios [45]. In contrast, deep learning-based networks (e.g., convolutional neural network (CNN) [46] and GNN [47]) have become the prevailing models for HSI classification (HSIC). In particular, two-dimensional (2-D) CNN [48] and three-dimensional (3-D) CNN [49] are designed for processing both spatial and spectral information. Although CNN performs well in HSIC, the contribution of feature maps generated by convolutional layers can vary. Consequently, recent advancements in attention mechanism-based techniques have emerged to address diverse feature maps and emphasize valuable features. In [50], an interleaved attention mechanism was designed for HSIC to enhance the interaction between spectral, spatial, and spectral–spatial information.

Nevertheless, the current methods for HSI cross-scene classification tasks encounter several challenges, outlined as follows.

1) Traditional DA feature extraction modules face challenges in handling variations in samples between SD and TD. It is necessary to enhance these modules by considering the impact of each sample on the model parameters. This enhancement allows for the extraction of more robust and transferable features.
2) Achieving a balance between marginal and conditional distributions presents challenges. The current DA methods effectively align two distributions between domains. However, fine-tuning the balance parameter between two distributions manually is a challenging task. This manual adjustment process can lead to the balance parameter being trapped in a local optimum, thereby limiting the overall performance of the adaptation process.

Addressing these challenges is crucial for advancing the field of HSI cross-scene classification. In this article, we propose a deep dynamic adaptation network based on joint correlation alignment (DDAN-JCA). DDAN-JCA mainly contains three modules: the dual-channel residual network (DCRN), the attention mechanism module (AMM), and the dynamic joint CORAL algorithm. The proposed DDAN-JCA method simultaneously aligns the marginal distribution of overall features extracted by DCRN and the conditional distribution of the local information extracted using AMM. At the same time, the dynamic joint CORAL algorithm can dynamically adjust the importance of marginal distribution and conditional distribution by using a balance factor. This article has three contributions.

1) To the best of the authors' knowledge, this is the first attempt to use joint CORAL distribution and dynamic distribution adaption for cross-scene HSI classification. The joint distribution can address the class information not considered by the marginal distribution to further improve classification accuracy.
2) The proposed DDAN-JCA can extract more marginal-invariant and conditional-invariant features, and we apply DCRN and AMM as feature extractors for deep DA. DCRN is used to efficiently extract the spatial–spectral joint features of HSI, avoiding other methods to extract single spectral or spatial information, and the obtained features are very suitable for marginal distribution alignment. AMM is used to pay attention to local information, and the obtained features are suitable for conditional distribution alignment.

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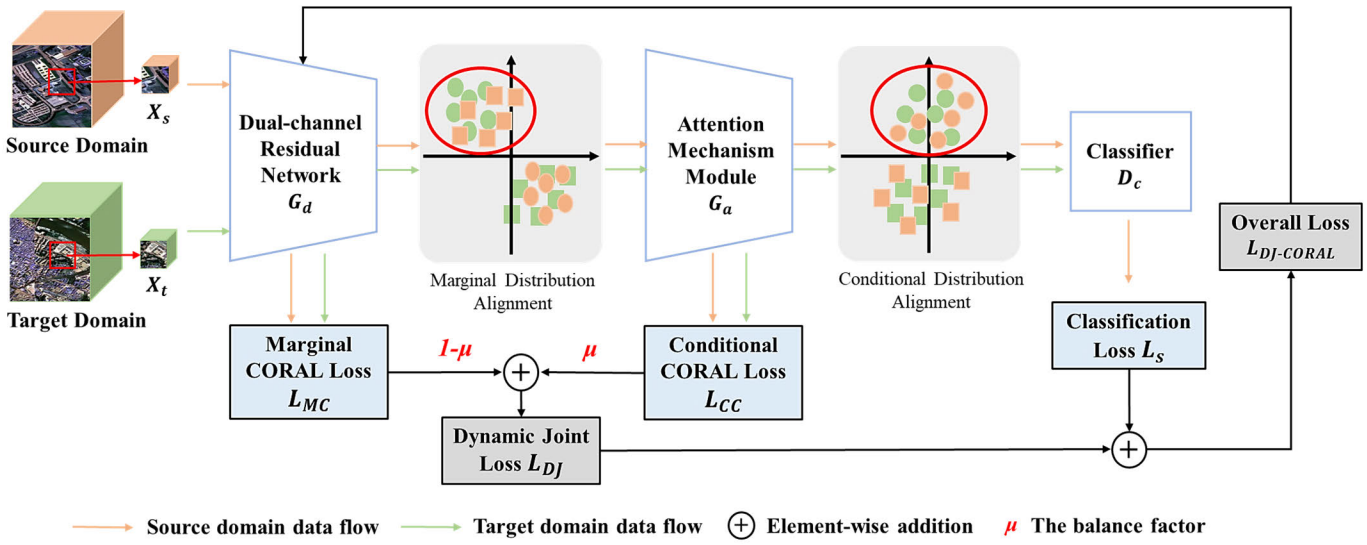


Fig. 1. DDAN-JCA.

3) We introduce a dynamic distribution adaptation strategy that dynamically adjusts the relative importance of marginal and conditional distributions. And, our proposed DDAN-JCA can find the optimal solution between the two distributions.

The remainder of this article is structured as follows. Section II provides a comprehensive explanation of the proposed DDAN-JCA method. Section III gives the experimental results obtained from applying DDAN-JCA to three publicly available HSI datasets. The performance evaluation and analysis are discussed in detail. Finally, a conclusion is provided in Section IV.

II. DDAN-JCA

The deep DA framework DDAN-JCA consists of the DCRN, the AMM, and the dynamic joint CORAL algorithm. Fig. 1 shows the methodology of the proposed DDAN-JCA, where the source data $X_s \in R^{n_s \times d}$ with labels $Y_s \in R^{n_s \times 1}$ and the target data $X_t \in R^{n_t \times d}$ without labels, d is the feature dimensionality, n_s and n_t represent the number of samples of source data and target data, respectively. First, the feature extractor G_d (DCRN) receives multiple 3-D cubes (select the 3-D HSI cube with a spatial dimension of 9 × 9), which are samples taken from the HSI of both SD and TD. To adjust the marginal distribution between domains in our method, we employ the marginal distribution CORAL strategy. Then, we utilize G_a (AMM) to capture local information effectively. And, we address the adaptation of conditional distribution between domains by incorporating the conditional distribution CORAL strategy. After that, we train a classifier (D_c) using the source-labeled data and employing two feature extractors (G_d and G_a) with shared weights to process source and target data. Finally, we dynamically adjust the losses associated with marginal and conditional distributions using the dynamic distribution adaptation strategy.

A. DCRN

Traditional CNNs have the problem of gradients disappearing as the number of layers increases, and residual networks can overcome this problem and extract deeper features. For cross-scene HSI classification tasks, traditional CNN-based feature extraction methods may overlook the rich spectral and spatial features of HSI data. Therefore, we employ the DCRN [41], which enables the effective fusion of spatial–spectral features, thus enhancing the extraction of discriminative features.

DCRN includes three key modules (see Fig. 2): 1) spectral feature extraction: this module focuses on extracting relevant spectral information from the input HSI data; 2) spatial feature extraction: this module is designed to capture spatial features by analyzing the spatial patterns present in the HSI data; and 3) spatial–spectral fusion: integrate the spatial and spectral features extracted from the previous modules to generate a comprehensive feature representation that includes both spectral and spatial features. To illustrate the proposed network structure, multiple 3-D cubes are fed into DCRN. Specifically, we consider 9 × 9 × 102 pixels cubes as an example. This network architecture enables the effective utilization of both spatial–spectral information within HSI data, facilitating cross-scene HSI classification tasks to extract discriminative features.

1) Spectral Feature Extraction: As depicted in Fig. 2, the spectral feature extraction channel consists of two convolutional layers and a residual block. In the first convolutional layer, we utilize 24 kernels with a size of 1 × 1 × 9. To enhance the focus on informative features, we employ a subsampling stride of (1, 1, 2), effectively by reducing redundant spectral features and emphasizing the most relevant ones. Consequently, the original input is transformed into 24 condensed 3-D cubes of size 9 × 9 × 47. Subsequently, the obtained 24 9 × 9 × 47 3-D cubes are passed through a residual block, which consists of two successive convolutional layers with 24 kernels of size 1 × 1 × 9. This residual block facilitates the extraction of more discriminative spectral features. After the residual block, the remaining cubes undergo convolution with 128 kernels of size 1 × 1 × 47. This operation produces 128 9 × 9 2-D spatial patches, where each pixel within a patch captures spectral information while retaining the original spatial features. By employing this spectral feature extraction channel, our method can effectively

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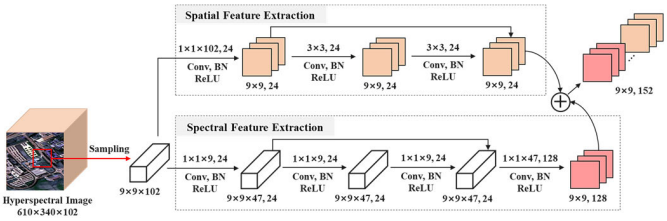


Fig. 2. DCRN.

utilize the rich spectral information present in the HSI data.

2) Spatial Feature Extraction: As depicted in Fig. 2, the spatial feature extraction channel includes a convolutional layer and a residual block. Initially, the input 3-D cubes are convolved with 24 kernels of size $1 \times 1 \times 102$. This convolutional operation aims to focus the network on detecting spatial correlations, enabling the extraction of informative spatial features. The resulting 24 9×9 patches are then passed through a residual block, which consists of two successive convolutional layers with 24 kernels of size 3×3. This residual block is designed to preserve and emphasize important spatial information, facilitating to obtain meaningful spatial features. By incorporating this spatial feature extraction channel, our method effectively utilizes the spatial features of HSI data.

3) Spatial–Spectral Fusion: The spatial–spectral fusion module is responsible for integrating information from spectral and spatial feature extraction channels. This fusion process combines the extracted spectral and spatial features in a meaningful manner. Specifically, the extracted spectral features, represented by 128 9×9 squares, are concatenated with the extracted spatial features, represented by 24 9×9 squares. By performing this concatenation operation, we obtain a total of 152 squares that are rich in both spatial–spectral information. This fusion of spatial and spectral features enables our model to utilize the complementary nature of these two types of information, leading to a more comprehensive and discriminative representation of the HSI data. The resulting fused features are subsequently utilized for downstream classification tasks, facilitating improved performance in cross-scene HSI classification.

B. AMM

To strengthen effective features and accelerate network convergence, Woo et al. [34] proposed an AMM consisting of spectral attention and spatial attention mechanisms and used it in DDAN-JCA. Fig. 3 displays the architecture of AMM employed in this article.

1) Spectral Attention Module: The spectral attention module employs an attention mechanism to reduce channel redundancy and capture the inte-channel relationships within the feature map. As illustrated in the upper part of Fig. 3, the feature map F, with dimensions $H \times W \times C$, serves as the input for the spectral attention module. Through a series of operations, a 1-D attention vector $M_{\text{spe}} \in R^{1\times1\times C}$, with dimensions $1 \times 1 \times C$, is generated. This attention vector captures the importance or relevance of each channel in the feature map. To refine the spectral information, the

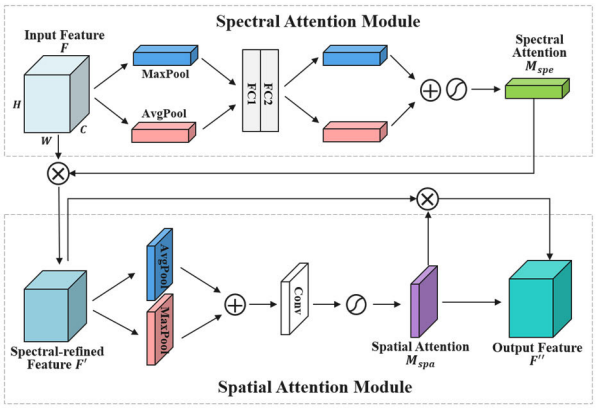


Fig. 3. AMM.

attention vector M_{spe} is applied to the feature map F using element-wise multiplication (denoted by $\otimes$). This operation results in the generation of the spectral-refined feature map F', which preserves the most salient and informative spectral features. By incorporating the spectral attention module, our method effectively highlights the most discriminative spectral information while reducing redundant channels, leading to enhanced feature representation and improved performance in cross-scene HSI classification tasks

$$M_{\text{spe}} = \sigma(\text{FC}(\text{MaxPool}(F))) + \sigma(\text{FC}(\text{AvgPool}(F))) \quad (1)$$

$$F' = F \otimes M_{\text{spe}}. \quad (2)$$

2) Spatial Attention Module: After the spectral attention module, the spatial attention module is applied to further enhance the informative regions within the feature map. As a result of the spatial attention module, a 2-D spatial attention map $M_{\text{spa}} \in R^{H\times W\times1}$ is generated. This map has dimensions $H \times W \times 1$, where H and W represent the height and width of the feature map, respectively. The spatial attention module highlights the spatial regions that contain the most relevant and discriminative information for the classification task

$$M_{\text{spa}} = \sigma\left(k^{9\times9}\left[\text{MaxPool}\left(F'\right); \text{AvgPool}\left(F'\right)\right]\right) \quad (3)$$

where $k^{9\times9}$ represents the convolution operation with a kernel size of 9×9.

Finally, the output feature map F'' is obtained by applying the attention mechanisms and convolutions in both the spectral and spatial attention modules. By incorporating both spectral and spatial attention mechanisms, our method effectively captures and emphasizes the most informative spectral and spatial features, leading to improved discriminative power and performance in cross-scene HSI classification tasks

$$F'' = F' \otimes M_{\text{spa}} \quad (4)$$

where F'' is the final feature map of F.

C. Dynamic Joint CORAL Algorithm

In the DDAN-JCA framework, we utilize the dynamic joint CORAL algorithm to align marginal and conditional distributions, respectively. In addition, to address the challenges of both global domain shifts and shifts within relevant subdomains, we introduce a dynamic joint CORAL algorithm. This

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algorithm dynamically adjusts the alignment strategy to effectively reduce distribution shifts in both the overall domain and specific subdomains. By combining these strategies, DDAN-JCA achieves comprehensive distribution adaptation, leading to improved cross-scene HSI classification performance.

1) Marginal Distribution Alignment: In our method, we employ CORAL [29] to align the marginal distributions in cross-scene HSI classification tasks. We aim to reduce the distribution discrepancy and enhance the transferability of the learned features in different scenes. We design the marginal CORAL loss as L_{MC}

$$L_{\mathrm{MC}} = \frac{1}{4d^2}\|C_s - C_t\|_F^2 \tag{5}$$

where $\|\cdot\|_F$ represents the matrix Frobenius norm, d is the feature dimensionality, and C_s and C_t refer to the covariance matrices of transformed features

$$C_s = \frac{1}{n_s - 1}\left(Z_s' Z_s - \frac{1}{n_s}\left(1'Z_s\right)'\left(1'Z_s\right)\right) \tag{6}$$

$$C_t = \frac{1}{n_t - 1}\left(Z_t' Z_t - \frac{1}{n_t}\left(1'Z_t\right)'\left(1'Z_t\right)\right) \tag{7}$$

where 1 represents a column vector with all elements being ones. n_s and n_t represent the total number of source and target data samples, respectively. Z_s and Z_t indicate the feature matrices of the source and target data. Z_s' and Z_t' are the transpose matrices of Z_s and Z_t, respectively.

2) Conditional Distribution Alignment: Referring to L_{MC}, we introduce the conditional CORAL loss (L_{CC}) in our method

$$L_{\mathrm{CC}} = \frac{1}{4d^2 C}\sum_{i=1}^{C}\left\|C_s^i - C_t^i\right\|_F^2 \tag{8}$$

$$C_s^i = \frac{1}{n_s^i - 1}\left(\left(Z_s^i\right)' Z_s^i - \frac{1}{n_s^i}\left(1'Z_s^i\right)'\left(1'Z_s^i\right)\right) \tag{9}$$

$$C_t^i = \frac{1}{n_t^i - 1}\left(\left(Z_t^i\right)' Z_t^i - \frac{1}{n_t^i}\left(1'Z_t^i\right)'\left(1'Z_t^i\right)\right) \tag{10}$$

where C represents the number of classes, and C_s^i and C_t^i represent the covariance matrices of the ith class in the SD and TD, respectively. n_s^i and n_t^i are the corresponding numbers of data samples for each class. $Z_s = [Z_s^1, Z_s^2, \ldots, Z_s^C] \in R^{n_s \times C}$ represents the feature matrix of the source data, where Z_s^i denotes the data of the ith class in the feature space. Similarly, the target data can be divided into C classes based on their pseudo-labels. $Z_t = [Z_t^1, Z_t^2, \ldots, Z_t^C] \in R^{n_t \times C}$ represents the target data that is predicted belong to each class.

3) Dynamical Distribution Adaptation Strategy: By jointly considering L_{MC} and L_{CC} with the balance factor $\mu \in [0, 1]$, we formulate the dynamic joint loss (L_{DJ}) as a weighted combination of the two loss terms. This dynamic joint loss allows us to effectively balance the contributions of marginal and conditional distribution alignments, enabling adaptive distribution adaptation for improved performance in cross-scene HSI classification tasks

$$L_{\mathrm{DJ}} = (1 - \mu)L_{\mathrm{MC}} + \mu L_{\mathrm{CC}} \tag{11}$$

where the parameter μ is employed to assign a relative weight to L_{MC} and L_{CC}. Taking inspiration from [35], we adopt an adaptive approach to dynamically estimate the value of μ for weight assignment

$$\mu = \frac{\sum_{i=1}^{C} d_A\left(Z_s^i, Z_t^i\right)}{d_A(Z_s, Z_t) + \sum_{i=1}^{C} d_A\left(Z_s^i, Z_t^i\right)} \tag{12}$$

where $d_A(\cdot, \cdot)$ means the A-distance [36].

D. Overall Loss Function

For DDAN-JCA, our training procedure includes two stages. First, we train a marginal distribution CORAL based on DCRN and define a loss function $\hat{L}$ that aims to minimize the discrepancy in second-order statistics between SD and TD. We have an effective alignment of the marginal distribution between domains, facilitating better feature extraction and transferability

$$\hat{L} = L_s + \alpha L_{\mathrm{MC}} \tag{13}$$

where α is a hyperparameter, and L_s is defined as the SD classification loss

$$L_s = \frac{1}{n_s} L_{ce}(f(Z_s), Y_s) \tag{14}$$

where $L_{ce}(\cdot, \cdot)$ represents the cross-entropy loss, and $f(\cdot)$ represents the domain-shared softmax classifier.

Second, we utilize the trained DCRN and AMM to predict pseudo labels for the target data. These pseudo-labels are then used in conjunction with the conditional distribution CORAL algorithm to further enhance the alignment between SD and TD. The overall loss function $L_{\text{DJ-CORAL}}$ consists of the dynamic joint loss and the classification loss, which enables the model to simultaneously optimize the domain alignment and classification accuracy, leading to improved cross-scene classification performance

$$L_{\text{DJ-CORAL}} = L_s + \alpha((1 - \mu)L_{\mathrm{MC}} + \mu L_{\mathrm{CC}}) \tag{15}$$

where α is used to balance the importance between classifier adaptation and distribution adjustment. And, μ is used to balance the contribution of L_{MC} and L_{CC}. By minimizing these losses, the model adapts to the TD while maintaining discriminative features and accurate classification.

III. Experiments

A. Experimental Settings

1) Experimental Datasets: To assess the efficacy of our method, we utilize three publicly available HSI datasets: Pavia, Houston, and Indiana.

a) Pavia dataset: The Pavia datasets consist of two subsets: the University of Pavia (UP, called SD) includes 610 × 610 pixels and 103 spectral bands, and the Pavia Center (PC, called TD) includes 1096 × 1096 pixels and 102 spectral bands. These datasets were captured using reflective optics system imaging spectrometer (ROSIS) sensors in Pavia, Northern Italy [37]. It should be noted that both datasets contain some invalid samples, which were excluded from the experiment. Consequently, the final UP and PC datasets consist of 610 × 315 and 1096 × 715 pixels, respectively. To ensure an equal number of spectral bands between UP

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TABLE I
PAVIA DATASET

Class		Number of samples	
ID	Name	Source (UP)	Target (PC)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584
Total		39332	39335

TABLE II
HOUSTON DATASET

Class		Number of samples	
ID	Name	Source (Houston 2013)	Target (Houston 2018)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

and PC and to have seven common categories, the last band in UP was removed, and their specific details can be found from Table I and Fig. 4.

b) Houston dataset: The Houston dataset consists of Houston 2013 [38] (called SD) and Houston 2018 [39] (called TD). The former includes 349 × 1905 pixels and 144 spectral bands. The latter includes 209 × 955 pixels and 48 spectral bands. These datasets were captured using different sensors and at different acquisition times in Houston, TX, USA. To align with the Houston 2018 dataset, we selected an overlapping region of 209 × 955 pixels from the Houston 2013 dataset and extracted 48 spectral bands from it. Both datasets include seven land cover classes, and their specific details can be found from Table II and Fig. 5.

c) Indiana dataset: The Indiana dataset was collected using airborne visible infrared imaging spectrometer (AVIRIS) in Northwest Tippecanoe Country, IN, USA. According to the method described in [40], the dataset was divided into two disjoint subsets, referred to as SD and TD. Each domain includes 400 × 300 pixels and 220 spectral bands. Both domains include seven land cover classes, and their specific details can be found from Table III and Fig. 6.

2) Compared Algorithms: We compared the proposed DDAN-JCA method with the machine learning method (support vector machines (SVMs) [42]) and deep learning methods, including DAN [27], domain adversarial neural network (DANN) [43], D-CORAL [30], CDA [44], and J-CORAL (DCRN and AMM with the joint CORAL algorithm). In all the experiments, a fair sampling strategy is

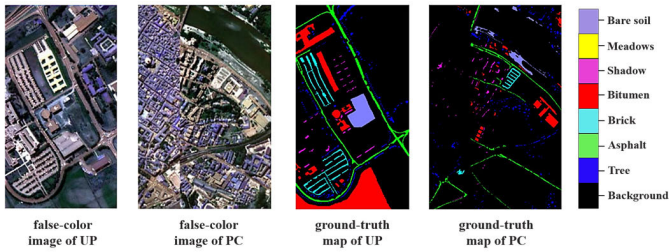


Fig. 4. Pavia dataset.

TABLE III
INDIANA DATASET

Class		Number of samples	
ID	Name	Source scene	Target scene
1	Concrete/Asphalt	4867	2942
2	Corn clean Till	9822	6029
3	Corn cleanTill EW	11414	7999
4	Orchard	5106	1562
5	Soybeans cleanTill	4731	4792
6	Soybeans cleanTill EW	2996	1638
7	Wheat	3223	10739
Total		42159	35701



Fig. 5. Houston dataset.

employed, where 10% of the samples are randomly selected from each class in the SD. This method ensures equal representation and fairness in the experimental setup. The methods in this article are implemented by pytorch. For the deep DA algorithms DAN, DANN, D-CORAL, CDA, J-CORAL, and DDAN-JCA, the patch_size is chosen from a set of options {5, 7, 9, 11}, while the batch_size is uniformly set to 64. The learning rate lr is selected from {0.001, 0.0001}. To assess the classification performance of each method, the evaluation metrics used are the overall accuracy (OA) and kappa coefficient (κ) [51] on the TD. These metrics provide a comprehensive measure of the accuracy and agreement between predicted and ground truth labels.

1) In the domain adaptation network framework, the feature extractor component utilizes the ResNet-50 architecture. The adaptation between domains is achieved by applying MMD. In all datasets, the DAN algorithm is executed for 1000 iterations, with a batch_size of 64 and a fixed learning rate of 0.001. These parameter settings ensure effective training and adaptation within the DAN framework.
2) In the DANN framework, the feature extractor component is implemented using the ResNet-18 architecture. The domain classifier, which is responsible for distinguishing between SD and TD, includes three fully

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TABLE IV
CLASSIFICATION ACCURACIES OF PC IN DIFFERENT METHODS

Class	SVM	DAN	DANN	D-CORAL	CDA	J-CORAL	Ours
1	93.57	96.72	96.49	92.83	93.29	91.29	**96.98**
2	88.76	73.76	89.91	85.70	93.18	93.44	**96.63**
3	67.87	70.74	75.55	**76.56**	76.07	74.02	73.57
4	28.35	77.05	57.59	59.99	65.72	67.34	**78.65**
5	99.68	**100.00**	**100.00**	99.79	99.93	99.93	**100.00**
6	46.10	54.57	71.79	72.09	**78.92**	75.42	78.56
7	79.18	67.26	68.03	76.54	69.41	78.86	**81.85**
OA(%)	64.14	77.91	79.87	80.12	82.35	83.49	**87.93**
κ	0.5488	0.7353	0.7578	0.7612	0.7880	0.8012	**0.8547**

TABLE V
CLASSIFICATION ACCURACIES OF HOUSTON 2018 IN DIFFERENT METHODS

Class	SVM	DAN	DANN	D-CORAL	CDA	J-CORAL	Ours
1	60.47	89.28	88.30	61.79	83.37	**91.06**	78.49
2	70.83	63.54	72.09	80.76	72.65	**84.78**	68.02
3	65.98	69.96	**73.07**	56.29	71.22	61.97	67.68
4	77.43	94.48	95.17	81.82	96.82	95.54	**100.00**
5	58.37	60.15	65.59	**94.69**	51.46	65.08	54.68
6	54.62	57.90	58.43	60.98	67.67	65.63	**81.71**
7	61.20	69.35	**70.84**	59.84	66.64	68.74	46.94
OA(%)	52.99	61.46	63.42	65.83	66.99	68.18	**72.78**
κ	0.4133	0.4610	0.4882	0.5125	0.5194	0.5384	**0.5656**

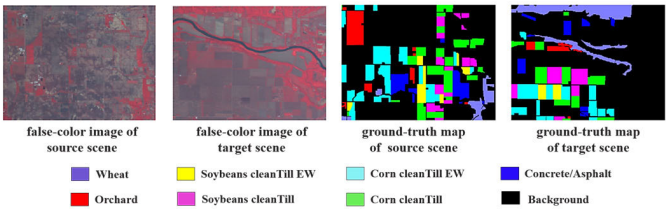


Fig. 6. Indiana dataset.

connected layers. For all datasets, the DANN algorithm is executed for 1000 iterations, with a batch_size of 64 and a fixed learning rate of 0.001. The input sizes of the three datasets are consistent with the specifications used in the DAN framework. These parameter configurations ensure effective feature extraction and domain discrimination within the DANN model.

3) In the D-CORAL framework, the experimental setup and hyperparameter settings for D-CORAL are identical to those described in [30] and are consistent with the ones used in this article.
4) In the class-wise distribution adaptation (CDA) framework, the learning rate lr is dynamically adjusted during training using the formula $\mathrm{lr} = \mathrm{lr}_0/(1+\alpha i)^p$, where α is set to 10, p is set to 0.75, and the training step i ranges from 0 to 1. The initial learning rate lr_0 is set to 0.001.
5) In the J-CORAL framework, the feature extractor component is implemented using the DCRN and AMM, and the adaptation between domains is achieved by applying joint CORAL algorithm. The learning rate lr is set to 0.0001, and the total number of training epochs is set to 100. The batch_size is set to 64. During the experiments, the parameter λ_1 is tested with ten different values: {0.1, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0}. Similarly, the parameter λ_2 is selected from the set {0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0}. These parameter values are evaluated to study their impact on the performance of J-CORAL during the experiments.
6) In the DDAN-JCA framework, the spatial size of the input cubes is set to 9×9 neighborhoods. lr is chosen as 0.0001, and the total number of training epochs is chosen as 100. Furthermore, the batch_size is chosen as 64. During the experiments, the parameter α is tested with five different values: {0.1, 0.5, 1.0, 1.5, 2.0}. These values are evaluated to assess their impact on the performance of the DDAN-JCA method.

B. Experimental Results

Tables IV–VI present the OA and κ of different methods on three datasets, and its classification accuracy for land cover classes. Considering the potential effects of random sampling, all experiments were conducted ten times. The final classification results are obtained by calculating the mean value from these ten runs, ensuring a reliable and robust evaluation of the methods' performance.

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TABLE VI
CLASSIFICATION ACCURACIES OF INDIANA (TD) IN DIFFERENT METHODS

Class	SVM	DAN	DANN	D-CORAL	CDA	J-CORAL	Ours
1	9.18	12.47	**15.64**	3.13	4.83	4.01	2.89
2	21.65	2.44	25.91	0.78	7.15	0.81	**32.86**
3	16.41	55.56	43.24	76.89	77.60	89.01	**92.23**
4	91.17	**95.26**	79.83	87.90	85.66	84.07	92.51
5	**2.46**	0.35	0.27	0.06	0.35	0.08	0.88
6	22.16	2.56	0.24	0	0.49	0	**45.60**
7	92.81	97.19	**98.90**	97.97	98.03	98.39	97.09
OA(%)	41.34	50.45	51.64	52.08	52.60	53.72	**56.07**
κ	0.2939	0.3428	0.3497	0.3931	0.3829	0.3969	**0.4171**

1) The deep learning algorithms exhibit superior classification performance compared with the traditional SVM method. For the Pavia dataset, the DAN algorithm achieves an OA that is 13.77% higher than the SVM method. Similarly, for the Houston dataset, the DAN algorithm achieves an OA that is 8.47% higher than the SVM method. Furthermore, for the Indiana dataset, the DAN algorithm is beyond the SVM method by 9.11% in terms of OA. This improvement can be attributed to the deep learning methods' ability to extract discriminative features through the establishment of hierarchical structured networks. The hierarchical structure enables deep learning algorithms to capture intricate patterns and nuances, resulting in enhanced classification performance.
2) When it comes to domain alignment or class alignment, relying solely on methods like MMD (DAN) or discriminant methods (DANN) or exclusively utilizing CORAL (D-CORAL) produces lower results compared to pseudo-labeling (CDA). The DAN and DANN methods focus on closing the distribution discrepancy between the two domains, ignoring its class information. In contrast, CDA trains SD comprehensively and predicts TD to obtain prediction probabilities. These probabilities are employed to establish alignment both within and between categories, taking into account both transferability and discriminability. In Table IV, the results on the Pavia dataset show that CDA outperforms DAN by 4.44%, DANN by 2.48%, and D-CORAL by 2.23%. Similarly, in Table V, the results on the Houston dataset indicate that CDA achieves an improvement of 5.53% over DAN, 3.57% over DANN, and 1.16% over D-CORAL. Finally, in Table VI, the results on the Indiana dataset demonstrate that CDA surpasses DAN by 2.15%, DANN by 0.96%, and D-CORAL by 0.52%. These findings highlight the advantage of CDA, which considers both transferability and discriminability, leading to improved performance on the evaluated datasets.
3) When considering both domain alignment and class alignment, the J-CORAL method outperforms CDA, which only focuses on class alignment. This can be observed from the results of the Pavia dataset in Table IV, where J-CORAL achieves a 1.14% higher accuracy compared to CDA. Similarly, in the Houston dataset (Table V), J-CORAL produces a 1.19% higher accuracy than CDA. In addition, in the Indiana dataset (Table VI), J-CORAL shows a 1.12% higher accuracy than CDA. These findings suggest that incorporating both marginal and conditional distributions can effectively enhance transferability while maintaining discrimination capabilities. J-CORAL utilizes the benefits of both types of alignments, leading to improved performance on the evaluated datasets.
4) By incorporating a dynamic distribution adaptation strategy, our method surpasses J-CORAL in performance. For the Pavia dataset (Table IV), our results exhibit a 4.44% higher accuracy than J-CORAL. Similarly, in the Houston dataset (Table V), our method achieves a 4.60% higher accuracy compared to J-CORAL. In addition, for the Indiana dataset (Table VI), our method produces a 2.35% higher accuracy than J-CORAL. The dynamic distribution adaptation strategy allows for the adjustment of the relative importance between marginal and conditional distributions, resulting in effective distribution adaptation. This flexibility helps us achieving superior performance in aligning distributions for improved classification results.

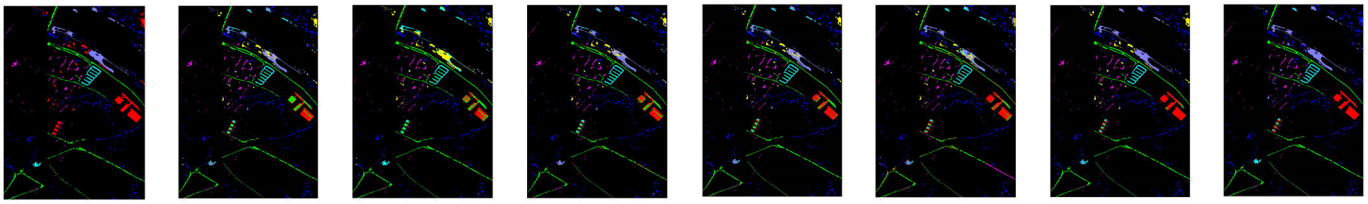
Furthermore, Figs. 7–9 depict the classification maps of PC, Houston 2018, and Indiana (TD). Our proposed DDAN-JCA method can accurately classify a larger number of samples. As a result, the generated mapping maps exhibit a higher level of consistency with ground-truth maps. The improved classification accuracy achieved by DDAN-JCA is visually evident in the prediction maps, indicating its effectiveness in capturing and preserving the underlying patterns and structures in the TD.

C. Discussion

In Section III-C, we will evaluate the effectiveness of DDAN-JCA in terms of both ablation analysis and parameters analysis.

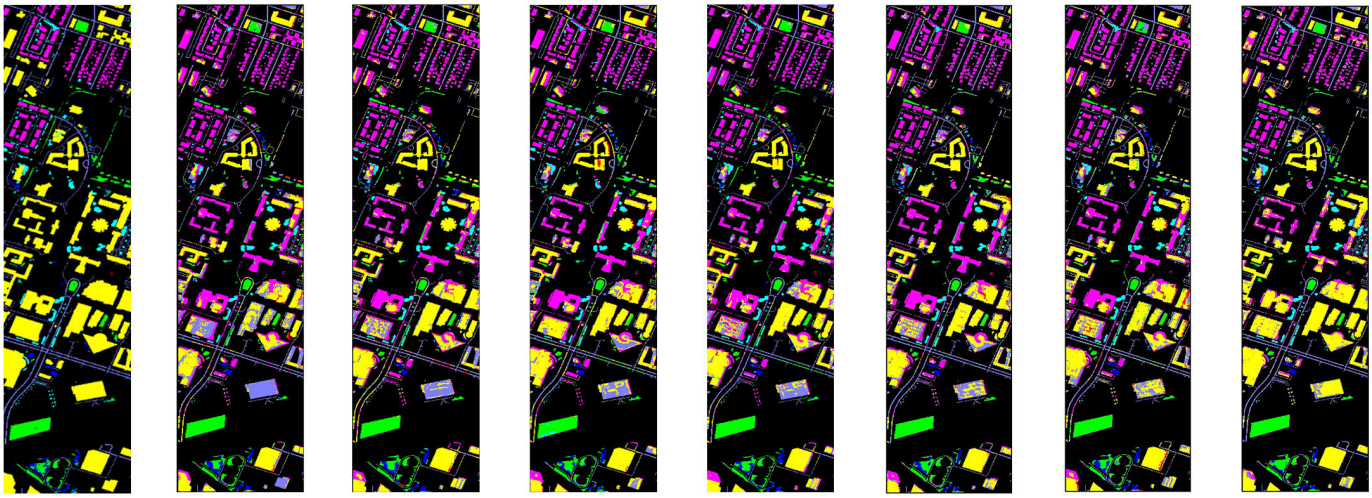
1) Ablation Analysis: To validate the effectiveness of each component in DDAN-JCA, we trained and evaluated

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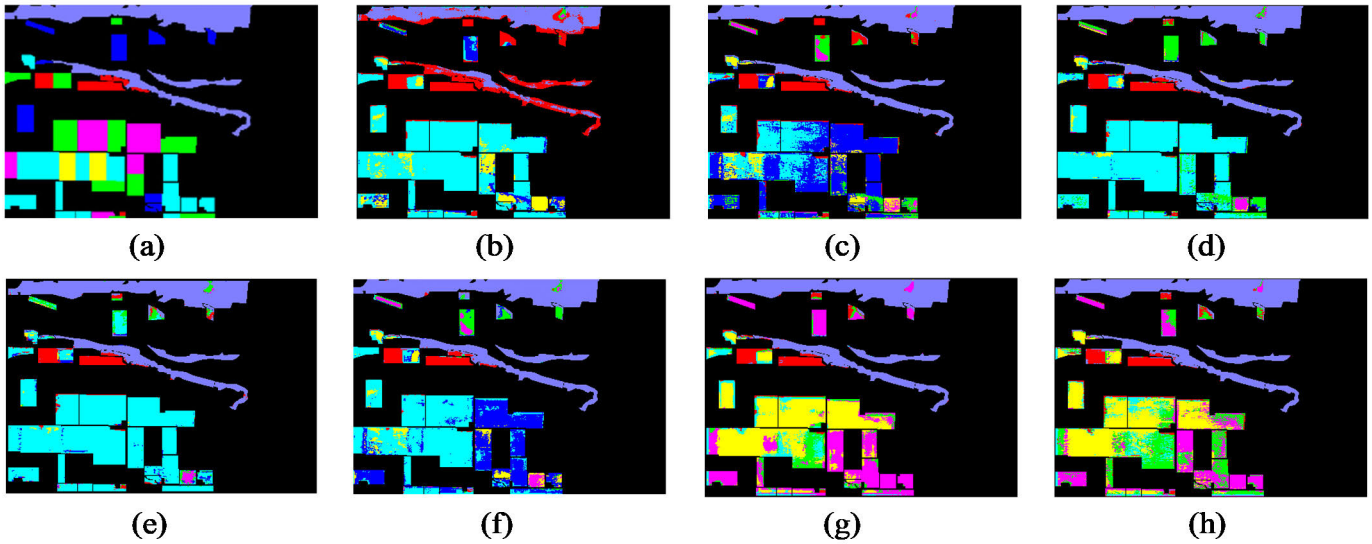
(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 7. Classification map of PC in different methods. (a) Ground truth. (b) SVM. (c) DAN. (d) DANN. (e) D-CORAL. (f) CDA. (g) J-CORAL. (h) DDAN-JCA (our method).



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 8. Classification map of Houston 2018 in different methods. (a) Ground truth. (b) SVM. (c) DAN. (d) DANN. (e) D-CORAL. (f) CDA. (g) J-CORAL. (h) DDAN-JCA (our method).



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 9. Classification map of Indiana (TD) in different methods. (a) Ground truth. (b) SVM. (c) DAN. (d) DANN. (e) D-CORAL. (f) CDA. (g) J-CORAL. (h) DDAN-JCA (our method).

different model variants using the same experimental parameters defined previously. As depicted in Table VII, these ablation experiments explore three different cases.

Case 1: Combination of DCRN and marginal CORAL algorithm to implement marginal distribution alignment of SD and TD.

Case 2: Combination of DCRN and AMM with the joint CORAL algorithm to implement marginal distribution alignment and conditional distribution alignment of SD and TD.

Case 3: Our method employs DCRN and AMM with the dynamic joint CORAL algorithm. The proposed DDAN-JCA method simultaneously aligns the marginal distribution of overall features extracted by DCRN and the conditional distribution of the local information extracted using AMM. At the same time, the dynamic joint CORAL algorithm can dynamically adjust the importance of marginal distribution and conditional distribution by using a balance factor.

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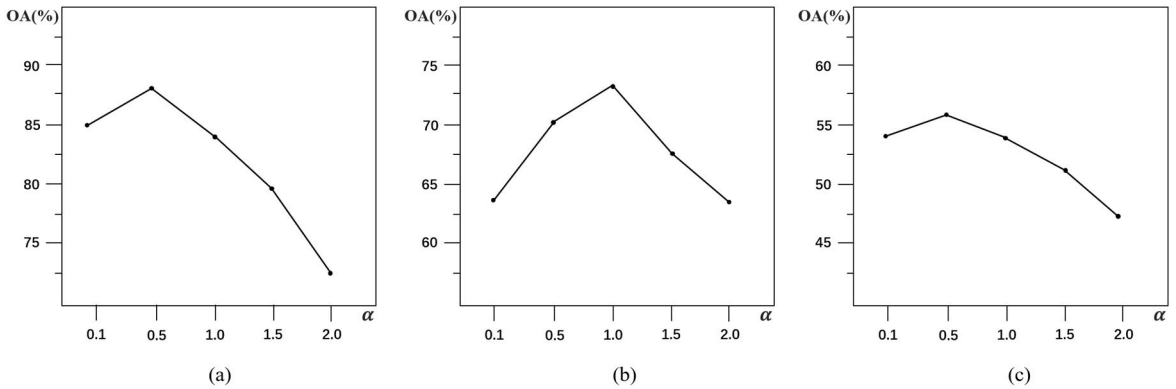


Fig. 10. Sensitivity analysis of α on three datasets. (a) PC. (b) Houston 2018. (c) Indiana (TD).

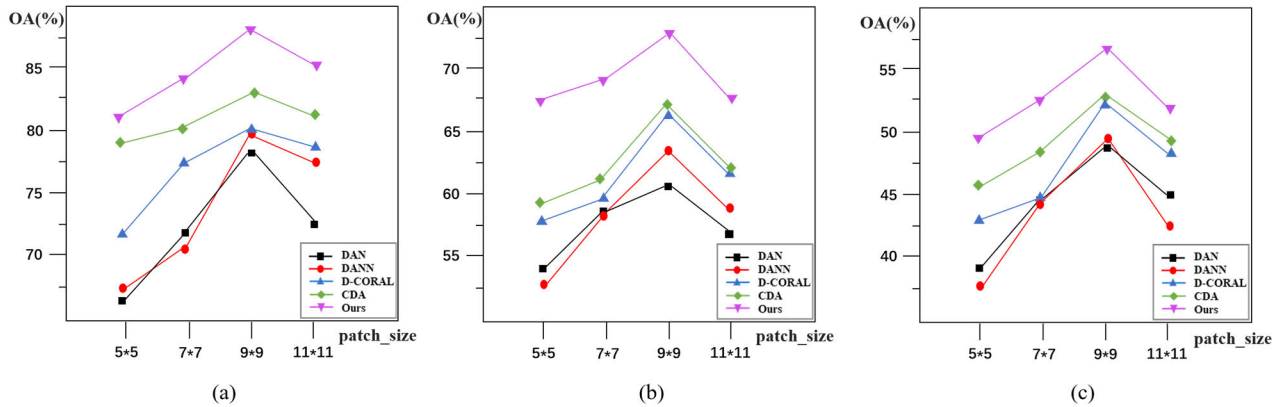


Fig. 11. Sensitivity analysis of patch_size on three datasets. (a) PC. (b) Houston 2018. (c) Indiana (TD).

TABLE VII
CLASSIFICATION RESULTS (OA%) ON THREE DIFFERENT SETTINGS

Task	Pavia	Houston	Indiana
case 1	80.12	65.83	52.18
case 2	83.49	68.18	53.72
case 3	87.93	72.78	56.07

TABLE VIII
CLASSIFICATION RESULTS (OA%) FOR L_{MC} AND L_{CC} OF DIFFERENT WEIGHT VALUES

Task	Pavia	Houston	Indiana
$\mu L_{MC} + (1-\mu)L_{CC}$	85.50	70.90	54.18
$(1-\mu)L_{MC} + \mu L_{CC}$	87.93	72.78	56.07

The experimental results substantiate the effectiveness of the proposed DDAN-JCA method.

2) Parameters Analysis: In the proposed DDAN-JCA method, several parameters need to be adjusted, including the balance factor μ, the parameter α, and the parameter patch_size. The balance factor μ plays an important role in determining the relative contributions of L_{MC} and L_{CC}. It helps striking a balance between aligning the overall distribution and adapting the classifier. The parameter α serves as a trade-off parameter, controlling the balance between classifier adaptation and distribution alignment. Finally, the parameter patch_size represents different patch sizes used for the input 3-D HSI cubes. The selection of appropriate patch_size can significantly impact the performance of the DDAN-JCA method.

a) Reliability of μ estimation: In this article, we adopt the adaptive method to calculate the balance factor μ using (12), instead of searching exhaustively in the interval $[0, 1]$ defined by it. In order to assess the reliability of such dynamic estimates and to determine whether (12) produces the optimal global results, we examine the weight values assigned to L_{MC} and L_{CC}. Table VIII reveals that μ is consistently greater than 0.5 in all three classification tasks. This analysis not only verifies the effectiveness of our adaptive estimation method but also suggests that assigning more weight to the conditional distribution loss can be beneficial in enhancing classification performance. It underscores the importance of dynamically adjusting μ to achieve an optimal balance between distribution alignment and classifier adaptation.

b) Sensitivity analysis of parameter α: Parameter α plays a critical role in determining the weight assigned to the dynamic joint loss. Assessing the impact of different values of α on the classification methods, we conducted a sensitivity analysis using five different values: {0.1, 0.5, 1.0, 1.5, 2.0}. For the PC dataset, shown in Fig. 10(a), the highest classification accuracy is achieved when α is set to 0.5. Similarly, for the Houston 2018 dataset, shown in

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Fig. 10(b), the optimal value for α is 1, resulting in the highest classification accuracy. Turning to the Indiana dataset (TD), illustrated in Fig. 10(c), an α value of 0.5 produces the best classification performance. This sensitivity analysis highlights the importance of appropriately selecting the value of α, as it significantly affects the performance of the classification methods on the three datasets.

c) Sensitivity analysis of parameter patch_size: To evaluate the impact of different patch sizes in different methods. We conducted a sensitivity analysis using patch_sizes of 5×5, 7×7, 9×9, and 11×11 for each method on the three datasets. Fig. 11 illustrates the results of this analysis. It can be observed that when using a patch_size of 9×9 on the Pavia, Houston, and Indiana datasets, different methods achieve the highest OA. This suggests that the choice of 9×9 patch_size used in the experiments was appropriate and produced favorable results for the performance of the different methods. It underscores the significance of carefully considering the patch_size during the preprocessing stage to achieve optimal classification outcomes.

IV. Conclusion

This article proposes an unsupervised DA framework DDAN-JCA for cross-scene HSI classification. Using the joint CORAL algorithm, DDAN-JCA can align both marginal and conditional distributions of SD and TD, and then, a balance factor μ is adopted to balance the relative importance of marginal and conditional distributions. In the experimental part, the effectiveness of DDAN-JCA was evaluated using three cross-scene hyperspectral datasets, the classification results and relevant analysis proved that DDAN-JCA has a powerful transferable capability.

References

[1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag. Replaces Newsletter*, vol. 5, no. 4, pp. 37–78, Dec. 2017, doi: 10.1109/MGRS.2017.2762087.

[2] S.-E. Qian, "Hyperspectral satellites, evolution, and development history," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7032–7056, Jun. 2021, doi: 10.1109/JSTARS.2021.3090256.

[3] J. Liu et al., "Estimating the forage neutral detergent fiber content of Alpine grassland in the Tibetan Plateau using hyperspectral data and machine learning algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4405017, doi: 10.1109/TGRS.2021.3105482.

[4] B. Luo, C. Yang, J. Chanussot, and L. Zhang, "Crop yield estimation based on unsupervised linear unmixing of multidate hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 162–173, Jan. 2013, doi: 10.1109/TGRS.2012.2198826.

[5] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag. Replaces Newsletter*, vol. 5, no. 1, pp. 8–32, Mar. 2017, doi: 10.1109/MGRS.2016.2616418.

[6] B. Deng, S. Jia, and D. Shi, "Deep metric learning-based feature embedding for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1422–1435, Feb. 2020, doi: 10.1109/TGRS.2019.2946318.

[7] C. I. Chang, *Hyperspectral Data Exploitation: Theory and Applications*, vol. 605. New York, NY, USA: Wiley-Interscience, Apr. 2007.

[8] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, Jun. 2023, doi: 10.1109/TNNLS.2021.3109872.

[9] W. Zhu, C. Zhao, S. Feng, and B. Qin, "Multiscale short and long range graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535815, doi: 10.1109/TGRS.2022.3199467.

[10] Y. Zhang, W. Li, R. Tao, J. Peng, Q. Du, and Z. Cai, "Cross-scene hyperspectral image classification with discriminative cooperative alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9646–9660, Nov. 2021, doi: 10.1109/TGRS.2020.3046756.

[11] J. Wang et al., "Generalizing to unseen domains: A survey on domain generalization," *IEEE Trans. Knowl. Data Eng.*, vol. 14, no. 8, pp. 8052–8072, Aug. 2015, doi: 10.1109/TKDE.2022.3178128.

[12] J. Peng, Y. Huang, W. Sun, N. Chen, Y. Ning, and Q. Du, "Domain adaptation in remote sensing image classification: A survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9842–9859, 2022, doi: 10.1109/JSTARS.2022.3220875.

[13] J. Peng, W. Sun, L. Ma, and Q. Du, "Discriminative transfer joint matching for domain adaptation in hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 6, pp. 972–976, Jun. 2019, doi: 10.1109/LGRS.2018.2889789.

[14] Y. Huang, J. Peng, Y. Ning, W. Sun, and Q. Du, "Graph embedding and distribution alignment for domain adaptation in hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7654–7666, Jul. 2021, doi: 10.1109/JSTARS.2021.3099805.

[15] Y. Huang, J. Peng, Y. Ning, Q. Cao, and W. Sun, "Distribution alignment and discriminative feature learning for domain adaptation in hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3128590.

[16] S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, "Domain adaptation via transfer component analysis," *IEEE Trans. Neural Netw.*, vol. 22, no. 2, pp. 199–210, Feb. 2011, doi: 10.1109/TNN.2010.2091281.

[17] M. Long, J. Wang, G. Ding, J. Sun, and P. S. Yu, "Transfer joint matching for unsupervised domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Columbus, OH, USA, Jun. 2014, pp. 1410–1417.

[18] J. Wang, Y. Chen, L. Hu, X. Peng, and P. S. Yu, "Stratified transfer learning for cross-domain activity recognition," in *Proc. IEEE Int. Conf. Pervasive Comput. Commun. (PerCom)*, Athens, Greece, Mar. 2018, pp. 1–10.

[19] J. C. Blitzer, R. T. Mcdonald, and F. C. N. Pereira, "Domain adaptation with structural correspondence learning," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, Sydney, NSW, Australia, Jul. 2006, pp. 120–128.

[20] Q. Gu, Z. Li, and J. Han, "Joint feature selection and subspace learning," in *Proc. 22nd Int. Joint Conf. Artif. Intell.*, Barcelona, Catalonia, Spain, Jul. 2011, pp. 1294–1299.

[21] J. Li, J. Zhao, and L. Ke, "Joint feature selection and structure preservation for domain adaptation," in *Proc. 25th Int. Joint Conf. Artif. Intell.*, New York, NY, USA, Jul. 2016, pp. 1697–1703.

[22] B. Fernando, A. Habrard, M. Sebban, and T. Tuytelaars, "Unsupervised visual domain adaptation using subspace alignment," in *Proc. IEEE Int. Conf. Comput. Vis.*, Sydney, NSW, Australia, Dec. 2013, pp. 2960–2967, doi: 10.1109/ICCV.2013.368.

[23] M. Baktashmotlagh, M. T. Harandi, B. C. Lovell, and M. Salzmann, "Domain adaptation on the statistical manifold," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Columbus, OH, USA, Jun. 2014, pp. 2481–2488, doi: 10.1109/CVPR.2014.318.

[24] J. E. Ball, D. T. Anderson, and C. S. Chan, "Comprehensive survey of deep learning in remote sensing: Theories, tools, and challenges for the community," *J. Appl. Remote Sens.*, vol. 11, no. 4, p. 1, Sep. 2017, Art. no. 042609.

[25] A. Farahani et al., "A brief review of domain adaptation," in *Proc. Adv. Data Sci. Inf. Eng., ICDATA IKE*, Oct. 2021, pp. 877–894.

[26] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. Smola, "A kernel two-sample test," *J. Mach. Learn. Res.*, vol. 13, pp. 723–773, Mar. 2012.

[27] M. Long and J. Wang, "Learning transferable features with deep adaptation networks," in *Proc. 32nd Int. Conf. Mach. Learn.*, Lille, France, Jul. 2015, pp. 97–105.

[28] Z. Li, X. Tang, W. Li, C. Wang, C. Liu, and J. He, "A two-stage deep domain adaptation method for hyperspectral image classification," *Remote Sens.*, vol. 12, no. 7, p. 1054, Mar. 2020, doi: 10.3390/rs12071054.

[29] B. Sun, J. Feng, and K. Saenko, "Return of frustratingly easy domain adaptation," in *Proc. 13th AAAI Conf. Artif. Intell.*, Phoenix, Arizona, Feb. 2016, pp. 2058–2065.

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[30] B. Sun and K. Saenko, "Deep CORAL: Correlation alignment for deep domain adaptation," in *Proc. Comput. Vis.-ECCV Workshops*, Amsterdam, The Netherlands, Cham, Switzerland: Springer, Oct. 2016, pp. 443–450.

[31] M. Long et al., "Deep transfer learning with joint adaptation networks," in *Proc. 34th Int. Conf. Mach. Learn.*, Sydney, NSW, Australia, Aug. 2017, pp. 2208–2217.

[32] W. Wang, L. Ma, M. Chen, and Q. Du, "Joint correlation alignment-based graph neural network for domain adaptation of multitemporal hyperspectral remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3170–3184, Mar. 2021, doi: 10.1109/JSTARS.2021.3063460.

[33] S. Zhu, B. Du, L. Zhang, and X. Li, "Attention-based multiscale residual adaptation network for cross-scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5400715, doi: 10.1109/TGRS.2021.3056624.

[34] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Munich, Germany, Sep. 2018, pp. 3–19.

[35] J. Wang et al., "Visual domain adaptation with manifold embedded distribution alignment," in *Proc. 26th ACM Int. Conf. Multimedia*, New York, NY, USA, Oct. 2018, pp. 402–410.

[36] B. Schölkopf, J. Platt, and T. Hofmann, "Analysis of representations for domain adaptation," in *Proc. Adv. Neural Inf. Process. Syst. Conf.* Cambridge, MA, USA: MIT Press, 2007, pp. 137–144.

[37] G. Licciardi et al., "Decision fusion for the classification of hyperspectral data: Outcome of the 2008 GRS-S data fusion contest," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 11, pp. 3857–3865, Nov. 2009, doi: 10.1109/TGRS.2009.2029340.

[38] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014, doi: 10.1109/JSTARS.2014.2305441.

[39] B. Le Saux, N. Yokoya, R. Hansch, and S. Prasad, "2018 IEEE GRSS data fusion contest: Multimodal land use classification [technical committees]," *IEEE Geosci. Remote Sens. Mag. Replaces Newsletter*, vol. 6, no. 1, pp. 52–54, Mar. 2018, doi: 10.1109/MGRS.2018.2798161.

[40] M. Ye, Y. Qian, J. Zhou, and Y. Y. Tang, "Dictionary learning-based feature-level domain adaptation for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1544–1562, Mar. 2017, doi: 10.1109/TGRS.2016.2627042.

[41] Y. Xu et al., "Dual-channel residual network for hyperspectral image classification with noisy labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502511, doi: 10.1109/TGRS.2021.3057689.

[42] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004, doi: 10.1109/TGRS.2004.831865.

[43] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 2030–2096, Jan. 2016.

[44] Z. Liu, L. Ma, and Q. Du, "Class-wise distribution adaptation for unsupervised classification of hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 508–521, Jan. 2021, doi: 10.1109/TGRS.2020.2997863.

[45] Q. Liu, J. Peng, G. Zhang, W. Sun, and Q. Du, "Deep contrastive learning network for small-sample hyperspectral image classification," *J. Remote Sens.*, vol. 3, p. 25, Jan. 2023.

[46] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5506812, doi: 10.1109/TGRS.2021.3093334.

[47] B. Xi et al., "Semisupervised cross-scale graph prototypical network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9337–9351, Mar. 2022, doi: 10.1109/TNNLS.2022.3158280.

[48] N. He et al., "Feature extraction with multiscale covariance maps for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 755–769, Feb. 2019, doi: 10.1109/TGRS.2018.2860464.

[49] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019, doi: 10.1109/TGRS.2019.2902568.

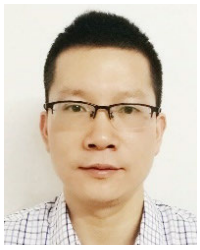
[50] Z. Deng et al., "A triple-path spectral–spatial network with interleave-attention for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5906–5923, 2022, doi: 10.1109/JSTARS.2022.3192470.

[51] Y. Ding, Y. Chong, S. Pan, and C. Zheng, "Diversity-connected graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518118, doi: 10.1109/TGRS.2023.3298848.



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